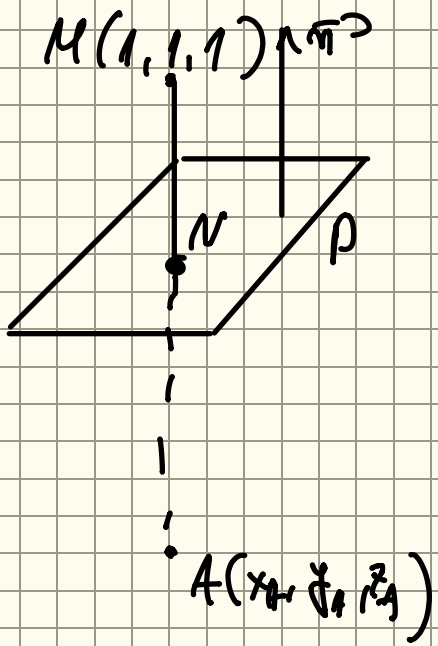
17) $M(1,1,1)$

$P: x+y-2z-6=0$



$\vec{n} = (1, 1, -2)$

$\left.\begin{matrix} \vec{n} \perp P \\ AM \perp P \end{matrix}\right| \Rightarrow \vec{n} \parallel AM$

$\Rightarrow AM: \begin{cases} x = 1+t \\ y = 1+t \\ z = 1-2t \end{cases}, t \in \mathbb{R}$

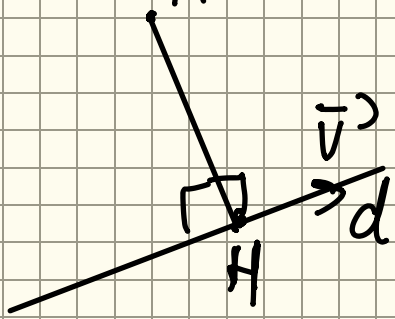
$N \in AM \Rightarrow N(1+t, 1+t, 1-2t), t \in \mathbb{R}$

$N \in P \Rightarrow 1+t+1+t-2+4t-6=0$

$\Rightarrow 6t = 6 \Rightarrow t = 1 \Rightarrow N(2,2,-1)$

$\Rightarrow A = 2N - M = (3,3,-3)$

18) $M(-1,-1,2)$, $d: \begin{cases} x = t+2 \\ y = 2t-1 \\ z = 3t+1 \end{cases}, t \in \mathbb{R}$



$\vec{v} = (1,2,3)$

$H = (t+2, 2t-1, 3t+1)$

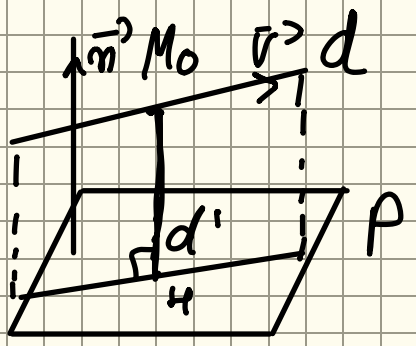
$\overrightarrow{MH} = (t+3, 2t, 3t-1)$

$\left.\begin{matrix} \vec{v} \parallel d \\ MH \perp d \end{matrix}\right| \Rightarrow \overrightarrow{MH} \perp \vec{v} \Rightarrow \overrightarrow{MH} \cdot \vec{v} = 0 \Rightarrow t+3+4t+9t-3=0$

$\Rightarrow 14t = 0 \Rightarrow t = 0 \Rightarrow H(2,-1,1)$

19) $d: \begin{cases} x = 2t+3 \\ y = -2t-1 \\ z = t+3 \end{cases}$

$P: x-3y+z+9=0$



$M_0 = (3, -1, 3) \in d$

$\vec{v} = (2, -2, 1)$

$\vec{n} = (1, -3, 1)$

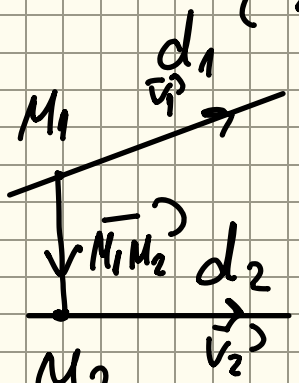
Let $M = d \cap P \Rightarrow 2t+3 + 6t+3 + t+3+9 = 0$

$\Leftrightarrow 9t + 18 = 0 \Rightarrow t = -2$

$\Rightarrow M(-1, 3, -1)$

???

10) $d_1 : \begin{cases} x = 3t - 7 \\ y = -2t + 4 \\ z = 3t + 4 \end{cases}$ $\quad d_2 : x - 1 = \frac{y+3}{2} = \frac{z+12}{-1}$



a) Suppose d_1 and d_2 coplanar

$\vec{M_1} = (-7, 4, 4)$

$\vec{M_2} = (1, -3, -12)$

$\vec{v_1} = (3, -2, 3)$

$\vec{v_2} = (1, 2, -1)$

$\overrightarrow{M_1M_2} = (8, -7, -16)$

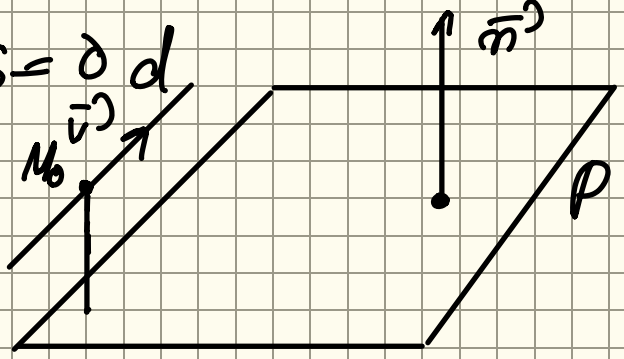
$\Rightarrow d_1, d_2$ coplanar $\Leftrightarrow \begin{vmatrix} 8 & -7 & -16 \\ 3 & -2 & 3 \\ 1 & 2 & -1 \end{vmatrix} = 0 \Leftrightarrow -202 \neq 0$

$\Rightarrow d_1, d_2$ are skew

b) $d(d_1, d_2) =$

11.2) a) $P: (\lambda-1)x - y + 3z - 5 = 0$ d

$$d: \begin{cases} x = 2t - 1 \\ y = -t + 2 \\ z = t - 2 \end{cases}$$



$M_0(-1, 2, 2)$

$\vec{v} = (2, -1, 1)$

$\vec{n} = (\lambda - 1, -1, 3)$

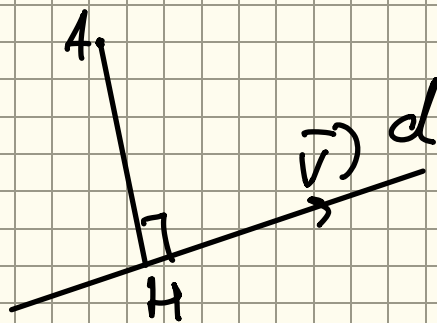
$\vec{n} \perp P \Big| \Rightarrow \vec{n} \cdot \vec{v} = 0$
$\vec{v} \parallel P$

$\Rightarrow 2\lambda - 2 + 1 + 3 = 0$

$\lambda = -1$

b) $dist(M_0, P) = \dfrac{|-2 \cdot (-1) - 1 \cdot 2 + 3 \cdot 2 - 5|}{\sqrt{(-2)^2 + (-1)^2 + 3^2}} = \dfrac{11}{\sqrt{14}}$

11.6) $A(1, -2, 4)$ $\quad d: \begin{cases} x = 2t + 1 \\ y = 2t + 2 \\ z = -t - 1 \end{cases}$



$M_0 = (1, 2, -1)$

$dist(A, d) = \dfrac{|\vec{v} \times \overrightarrow{M_0A}|}{|\vec{v}|}$

$\vec{v} = (2, 2, -1)$

$\overrightarrow{M_0A} = (0, -4, 5)$

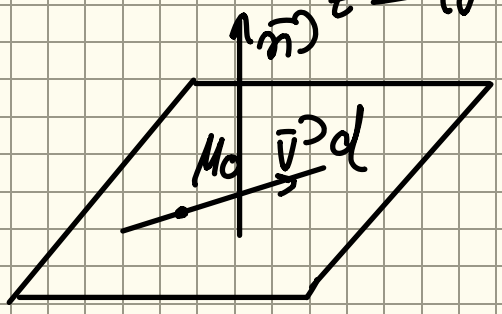
$$\vec{v} \times \overrightarrow{M_0A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 2 & -1 \\ 0 & -4 & 5 \end{vmatrix} = 6\vec{i} - 10\vec{j} - 8\vec{k}$$

$\Rightarrow |\vec{v} \times \overrightarrow{M_0A}| = \sqrt{36 + 100 + 64}$

$= 10\sqrt{2}$

$|\vec{v}| = \sqrt{4 + 4 + 1} = 3 \Rightarrow dist(A, d) = \dfrac{10\sqrt{2}}{3}$

11 S) $d: \begin{cases} x = t + \beta \\ y = 4t + 1 \\ z = (\alpha - 1)t - 2 \end{cases}$ $\quad P: 2x - 3y + 5z + 13 = 0$



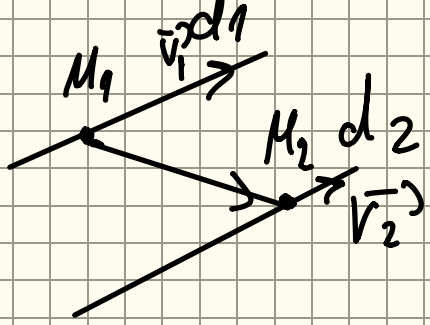
$\vec{n} = (2, -3, 5)$
$\vec{v} = (1, 4, \alpha - 1)$
$M_0(\beta, 1, -2)$

$M_0 \in P \Rightarrow 2\beta - 3 - 10 + 13 = 0$
$\beta = 0$

$\vec{n} \perp P \;\Big|\; \Rightarrow \vec{n} \perp \vec{v} \Leftrightarrow \vec{n} \cdot \vec{v} = 0 \Rightarrow$
$\vec{v} \parallel P$

$2 - 12 + 5\alpha - 5 = 0 \Rightarrow \alpha = 3$

11 F) $d_1: \frac{x-3}{2} = \frac{y-4}{-1} = \frac{z-5}{1}$ $\quad d_2: \begin{cases} x = 3 - 2t \\ y = 1 + t \\ z = 1 - t \end{cases}$



$M_1(3, 4, 5)$
$M_2(3, 1, 1)$
$\vec{v_1} = (2, -1, 1)$
$\vec{v_2} = (-2, 1, -1)$

$\vec{v_1} = -1 \vec{v_2} \Rightarrow \vec{v_1} \parallel \vec{v_2}$

$$\text{dist}(d_1, d_2) = \frac{|\vec{v_2} \times \vec{M_1 M_2}|}{|\vec{v_2}|}$$

$\overrightarrow{M_1M_2} = (0, 3, 4)$

$$\vec{v_2} \times \overrightarrow{M_1M_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 3 & 4 \\ -2 & 1 & -1 \end{vmatrix} = -7\vec{i} + 8\vec{j} + 6\vec{k}$$

$|\vec{v_2} \times \overrightarrow{M_2M_1}| = \sqrt{49+64+36} = \sqrt{149}$

$|\vec{v_2}| = \sqrt{6} \Rightarrow d_{int} = \sqrt{\frac{149}{6}}$